ADA University, Fall 2023, MATH 1101 Discrete Structures/Fund of Math

SAMPLE TASKS Midterm 1 (October 14, 2021). **SOLUTION SHEET**

25 points

- Special sequences for the case of THREE input variables A, B, C (or x, y , z):

  A (or x) = 00001111, B (or y) = 00110011, C (or z) = 01010101

- The truth tables for the minterms (**using the special sequences for x, y, z**) are as follows:

Table 1

Representation of minterms as sequences of bits (based on the special sequences for x, y, z)

xyz=00000001	xyz′=00000010	xy′z=00000100	xy′z′=00001000
x′yz=00010000	x′yz′=00100000	x′y′z=01000000	x′y′z′=10000000

SOLUTIONS

Task 1 (4=1+1+1+1 pt). Design a three-input **minimal** AND-OR circuit L with the following truth table:

T =[x, y, z; L]=[00001111, 00110011, 01010101; **01011111**]

Solution.

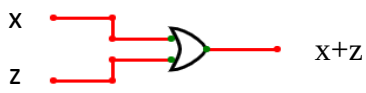
Step 1. (1 pt) Input sequences for the given circuit are special sequences. Represent the output sequence of bits generated by the given circuit, namely, **01011111** as a sum of minterms according to the Table 1. We have **01011111**=x′y′z+x′yz+xy′z′+xy′z+xyz′+xyz

Step 2. (1 pt) Put checks in K-map according to the last expression

(A)	yz	yz′	y′z′	y′z
x	V	V	V	V
x′	V			V

Step 3. (1 pt) MSoP=x+z

Step 4. (1 pt) **Minimal** AND-OR circuit L which corresponds to is MSoP



Task 2 (4=2+1+1 pt). Find the truth table T=T(E) for the Boolean expression

E=E(x, y, z)=(x′y′+z)′yz′+(x′z+y)′(y′z+x)+x′z+xy′z′

Hint.

Step 1. Find SoP form for E. (**2 pt**)

Step 2. Find CSoP form (**1pt**)

Step 3. Use Table 1 to find T(E) (T(E) will be the sequence of eight digits 0 or 1). (**1 pt**)

Solution.

Step 1. (2 pt). Find SoP form for E.

E(x, y, z)= (x′y′+z)′yz′+(x′z+y)′(y′z+x)+x′z+xy′z′=

=[(x′y′)′z′)]yz′+[(x′z)′y′](y′z)+x)+x′z+xy′z′=

=[(x+y)z′)]yz′+[(x+z′)y′](y′z)+x)+x′z+xy′z′ =

=xyz′+yz′ + (xy′y′z +z′y′y′z+ xy′x+ z′y′x +x′z+xy′z′ =

=xyz′+yz′+xy′z +xy′+xy′z′+x′z=(apply absorption law)=yz′+xy′+x′z=yz′+xy′+x′z

Step 2. (1 pt). Find CSoP form

E(x, y, z)=yz′+xy′+x′z=1yz′+xy′1+x′1z =(xyz′+x′yz′)+(xy′z+xy′z′)+(x′yz+x′y′z)=

=xyz′+x′yz′+xy′z+xy′z′+x′yz+x′y′z

Step 3. (1 pt). Find T(E)

T(E)=xyz′+x′yz′+xy′z+xy′z′+x′yz+x′y′z = 0000 0010 + 0010 0000 + 0000 0100 +

+ 0000 1000 + 0001 0000 + 0100 0000 = 0111 1110

Answer: T(E)= **0111 1110**

Task 3 (4 pt=1+3)

Find a Boolean expression E=E(x, y, z) in **MSoP form** corresponding to the truth table T(E):

T(E)=1101 1100;

Hint. In each case you must

1. Find CSoP form for E by using Table 1. (**1 pt**)
2. Then use Karnaugh map method to reduce CSoP form to an MSoP form. If there exist two or more MSoP forms it will be enough to find only one MSoP form. (**3 pt**)

Solution.

Step 1. (1 pt). Find CSoP form for E by using Table 1.

CSoP(E)=x′y′z′+x′y′z+ x′yz + xy′z′+xy′z

Step 2. (3 pt). Use Karnaugh map method to reduce CSoP form to an MSoP form. According to terms in the formula for CSoP put checks into corresponding cells:

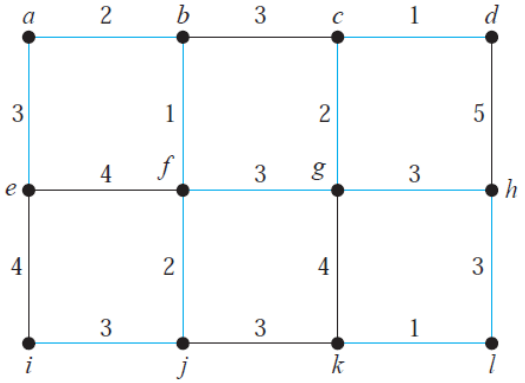
(A)	yz	yz′	y′z′	y′z
x			V	V
x′	V		V	V

It easy to see that **MSoP(E)=y′+x′z** **(Answer)**

Task 4 (3 pt)

Use Prim's algorithm to find a minimum spanning tree in the graph.

Solution. Minimum spanning tree constructed using Prim's algorithm is shown in Figure 1. The successive edges chosen are displayed.



(a)

Choice	Edge	Weight
1	{b, f}	1
2	{a, b}	2
3	{f, j}	2
4	{a, e}	3
5	{i, j}	3
6	{f, g}	3
7	{c, g}	2
8	{c, d}	1
9	{g, h}	3
10	{h, l}	3
11	{k, l}	1
	Total:	24

(b)

FIGURE 1 A Minimum Spanning Tree Produced Using Prim's Algorithm.

Successively add edges with minimum weight that do not form a simple circuit with those edges already chosen. Stop after (n-1) edges have been selected.
The edges in color in Figure 1 show a minimum spanning tree produced by Prim's algorithm, with the choice made at each step displayed. ■

Task 5 (4=2+2 pt)

1. Find a MSoP for the Boolean expression E given by Table (A) (*Write as an expression*). **(2pt)**
2. Find a MSoP for the Boolean expression E given by Table (B). (*Write as an expression*). (**2pt**)

(A)	zt	zt′	z′t′	z′t
xy	V	V		V
xy′			V	V
x′y′	V	V		
x′y	V	V	V	V

(B)	zt	zt′	z′t′	z′t
xy	V			V
xy′		V	V	V
x′y′	V	V	V	V
x′y	V	V		

Solution.

Subtask (A).

MSoP(A)=**x′y + x′z + yz + yt +xy′z′** (E_S=5, E_L=12)

Subtask (B).

MSoP(B)=$\mathbf{x'z + y't' + y'z' + xyt}$ (E_S=4, E_L=9)

Task 6 (5=1+2+2 pt)

Subtask 6.1 (1 pt). Find the negation of the proposition below

$\forall x \exists y \forall z P(x, y, z)$

Subtask 6.2 (2 pt). Domain of all predicates P(x) below is the set of *all integers*. What are the truth values of the corresponding propositions?

a) $\exists !x(x>1)$ **(0.5 pt)** b) $\exists !x(x^2=1)$ **(0.5 pt)**

c) $\exists !x(x+3=2x)$ **(0.5 pt)** d) $\exists !x(x=x+1)$ **(0.5 pt)**

Solution.

Subtask 6.1 (1 pt). $\neg[\forall x \exists y \forall z P(x, y, z)] = \exists x(\neg[\exists y \forall z P(x, y, z)]) =$

$= \exists x(\forall y(\neg[\forall z P(x, y, z)])) = \exists x(\forall y(\exists z[\neg P(x, y, z)]))$;

Subtask 6.2 (2 pt).

a) False **(0.5 pt)** b) False **(0.5 pt)** c) True **(0.5 pt)** d) False **(0.5 pt)**

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